

1

Public Dataset

EPA/UCI
PM_{2.5}, O₃, NO₂

2

Preprocessing

Normalize - Clean
Feature Engineering

3

Model Training

Linear Regression
Random Forest
Gradient Boosting
Tiny LSTM

4

Edge Simulation

CPU Throttle
RAM Limit

Prediction Accuracy

RMSE - MAE - R²
Compare across all models

RQ1

Computational Efficiency

Inference time - Memory usage
Model size (MB)

RQ2

Deployment Trade Off

Accuracy vs speed vs size
Edge vs cloud comparison

RQ3

Findings & Guidance

Best model for edge deployment
When edge outperforms cloud
Lightweight model recommendations